MATH 3551 A — Abstract Algebra I
Practice Midterm 2 — Version 2 — SOLUTIONS

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Version 2 — September 28, 2026. Generated by GPT-5.6 Sol in the Codex desktop harness. Please notify Professor Dupuy if something looks off.

What changed from Version 1. The October 26 revision, drafted with Codex, moves §§3.4–3.5 and Chapter 4 to Midterm 3. This version now stops at §3.3. Material deferred from Midterm 1 — §1.7, §2.2, and §§3.1–3.2 — remains included. The October 20 revision shortened the exam and rescored the retained parts. Version 1 remains available on the course webpage so that the changes can be compared directly.

This practice exam targets about twenty minutes of work in a fifty-minute period. Please report any error, ambiguity, out-of-coverage question, or item that takes much longer than its point value suggests.

Problem 1. (20 points) **True or false; prove or disprove.**

(a) **TRUE.** This is part of the Second Isomorphism Theorem: since $N \trianglelefteq G$ we have $HN = NH$, and a product of subgroups is a subgroup exactly when the two commute setwise.

(b) **FALSE.** In $G = D_8 = \langle r, s \mid r^4 = s^2 = 1,\ rs = sr^{-1} \rangle$, take $H = \langle r^2, s \rangle$ and $K = \langle s \rangle$. The group H has index 2 in G, hence is normal, and H is abelian, so $K \trianglelefteq H$. But $rsr^{-1} = sr^2 \notin K$, so K is not normal in G.

(c) **TRUE.** A subgroup of index 2 has only one left coset outside itself and one right coset outside itself; each equals its complement, so left and right cosets agree.

(d) **FALSE.** Taking $N = 1$ in the nonabelian group S_3 gives $S_3/N \cong S_3$, a nonabelian quotient.

Grading note. Most of the credit here is in the *reason*, and the reason must be specific: a named counterexample, the actual permutation, or the actual count. A circled answer with no reason earns two of the five points; a wrong circle with a reason that is right about the underlying fact earns the three reason-points. Give full credit for a correct counterexample even if it is not the one printed above. The parenthetical remarks in each answer flag the error to expect on that item.

Problem 2. (30 points) **A quotient of $\mathbb{Z}/30\mathbb{Z}$.**

(a) $H = \{\overline{0}, \overline{6}, \overline{12}, \overline{18}, \overline{24}\}$ and $N = \{\overline{0}, \overline{10}, \overline{20}\}$. Their intersection is $\{\overline{0}\}$, and $H + N = \langle\overline{2}\rangle$, the order-15 subgroup of G.

(b) $H/(H \cap N) = H \cong \mathbb{Z}/5\mathbb{Z}$. The quotient $(H + N)/N$ has order $15/3 = 5$ and is cyclic, hence is also $\mathbb{Z}/5\mathbb{Z}$. Explicitly $h + (H \cap N) \mapsto h + N$ is well defined, injective because its kernel is $H \cap N$, and surjective onto $(H + N)/N$.

Grading note. In (a), both subgroups and the intersection must be computed. In (b), identifying two order-five groups without the induced map is incomplete verification of the theorem.

Problem 3. (30 points) **Subgroups above N.**

(a) The subgroups containing N are $N = \langle\overline{6}\rangle$ (order 3), $\langle\overline{3}\rangle$ (order 6), $\langle\overline{2}\rangle$ (order 9), and $G = \langle\overline{1}\rangle$ (order 18). A cyclic group has one subgroup of each order dividing 18, and these are exactly the orders divisible by 3. Thus the list is complete.

(b) G/N has order $18/3 = 6$ and is cyclic, so $G/N \cong \mathbb{Z}/6\mathbb{Z}$. The four groups H/N above have respective orders $1, 2, 3, 6$; they are the unique subgroups of $\mathbb{Z}/6\mathbb{Z}$ of those orders. This is the Fourth Isomorphism Theorem correspondence.

Grading note. Require all four containing subgroups, including N and G. The matched quotient orders must be $1, 2, 3, 6$.

Problem 4. (20 points) **A normalizer in S_3.**

(a) An element commuting with $(1\,2)$ must preserve its fixed point 3 and the pair $\{1,2\}$. Thus $C_{S_3}(H) = H = \{1, (1\,2)\}$.

(b) Since H has one nonidentity element, an element normalizes H exactly when it centralizes $(1\,2)$. Hence $N_{S_3}(H) = H$, which is proper in S_3, so H is not normal.

Grading note. Distinguish the centralizer and normalizer definitions, even though they agree for this order-two subgroup.